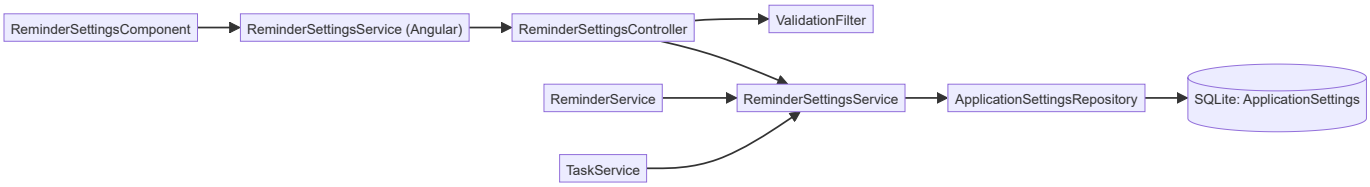
Reminder Settings – Technical Documentation

> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System **Parent Module:** Settings · **Child Module:** 08.2 Reminder Settings **Version:** 1.0 · **Last Updated:** 31-Oct-2026 **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core) **Author:** Jojo George (jojogeorge3344@gmail.com) © 2026 Jojo George. All rights reserved.

1. Technical Overview

Reminder Settings is `ReminderSettingsController` (class-level `settings.reminders.manage`) over `ReminderSettingsService` and the shared `ApplicationSettingsRepository`. It reads and writes five `Reminder.*` keys and `Notification.MailEnabled` in the `ApplicationSettings` table, and returns the application timezone read-only. The reminder engine (`ReminderService`, Communication module) and `TaskService` (Monitoring module) consume these values.

2. Architecture Flow



3. Backend File Structure

Layer	Path	Files
Controller	`TaskFlow.Api/Controllers/Settings/`	`ReminderSettingsController.cs`
Service	`TaskFlow.Api/Services/Settings/ReminderSettings/`	`IReminderSettingsService.cs`, `ReminderSettingsService.cs`
Repository (shared with 08.1)	`TaskFlow.Api/Repositories/Settings/ApplicationSettings/`	`IApplicationSettingsRepository.cs`, `ApplicationSettingsRepository.cs`
DTOs	`TaskFlow.Api/Dtos/Settings/ReminderSettings/`	`ReminderSettingsDtos.cs` (`ReminderSettingsDto`, `UpdateReminderSettingsRequest`)
Validators	`TaskFlow.Api/Dtos/Settings/ReminderSettings/Validators/`	`UpdateReminderSettingsRequestValidator.cs`
Entities	`TaskFlow.Api/Models/Entities/`	`ApplicationSetting.cs`
Enums	`TaskFlow.Api/Models/Enums/`	`ReminderFrequency.cs`
Constants	`TaskFlow.Api/Common/Constants/`	`SettingKeys.cs`
Seeders	`TaskFlow.Api/Persistence/Seeders/`	`SettingsInitializer.cs`
Config	`TaskFlow.Api/Config/`	`ApplicationOptions.cs` (timezone fallback)

Layer	Path	Files
Migrations	`TaskFlow.Api/Persistence/Migrations/`	`20261013104629_AddTaskReminderSettings.cs` (per-task reminder columns on `Tasks`)

4. Frontend File Structure

Type	Path	Purpose
Page component	`src/app/modules/settings/reminder-settings/pages/reminder-settings.component.{ts,html}`	`ReminderSettingsComponent` (Screen 13)
Models	`src/app/modules/settings/reminder-settings/pages/reminder-settings.model.ts`, `src/app/modules/settings/models/settings.model.ts`	`ReminderSettings` interface
API service	`src/app/modules/settings/reminder-settings/services/reminder-settings.service.ts`	`ReminderSettingsService.get()/save()`
Shared options	`src/app/shared/utils/reminder-options.ts`	`FIRST_REMINDER_OPTIONS`, `SECOND_REMINDER_OPTIONS`, `OVERDUE_FREQUENCY_OPTIONS`, `withReminderDay()`
Route / Guard	`src/app/modules/settings/settings.routes.ts` (`settings/reminders`), `src/app/shared/auth/auth.guard.ts`	`permissionGuard` with `SettingsReminders`
Menu	`src/app/layouts/nav-items.ts`	"Reminder Settings" item when `SettingsReminders` is held

5. API & Logic

5.1 API Endpoints

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
1	GET	`/api/v1/settings/reminders`	Current reminder configuration plus read-only timezone	`settings.reminders.manage` (class level)	–	`ReminderSettingsDto`
2	PUT	`/api/v1/settings/reminders`	Save reminder configuration (no timezone field)	`settings.reminders.manage` (class level)	`UpdateReminderSettingsRequest`	`ReminderSettingsDto`

5.2 Request / Response Samples

```
// PUT /api/v1/settings/reminders
{ "enabled": true, "deadlineDayEnabled": true, "defaultFirstDays": 2, "defaultSecondDays": 1,
"defaultOverdueFrequency": "Daily", "mailEnabled": true }
```

```
// 200 OK
{ "enabled": true, "deadlineDayEnabled": true, "defaultFirstDays": 2, "defaultSecondDays": 1,
"defaultOverdueFrequency": "Daily", "mailEnabled": true, "timezone": "Asia/Kolkata" }
```

5.3 Service Layer Logic

Method: `ReminderSettingsService.GetAsync(CancellationToken)`

1. Reads `Application.Timezone`, `Reminder.Enabled`, `Reminder.DeadlineDayEnabled`, `Reminder.DefaultFirstDays`, `Reminder.DefaultSecondDays`, `Reminder.DefaultOverdueFrequency` and `Notification.MailEnabled`.
2. Falls back to the defaults `true`, `true`, `2`, `1`, `Daily`, `true`. The timezone falls back to `ApplicationOptions.Timezone`.

Method: `ReminderSettingsService.UpdateAsync(int userId, UpdateReminderSettingsRequest request, CancellationToken)`

1. Throws `ArgumentException("Every reminder setting must be provided.")` if any field is null.
2. Upserts the five `Reminder` keys (category `Reminder`) and `Notification.MailEnabled` (category `Notification`). It sets `UpdatedByUserId` and `UpdatedAt`. Ints use invariant culture.
3. Saves and returns `GetAsync()`. `Application.Timezone` is never written here.

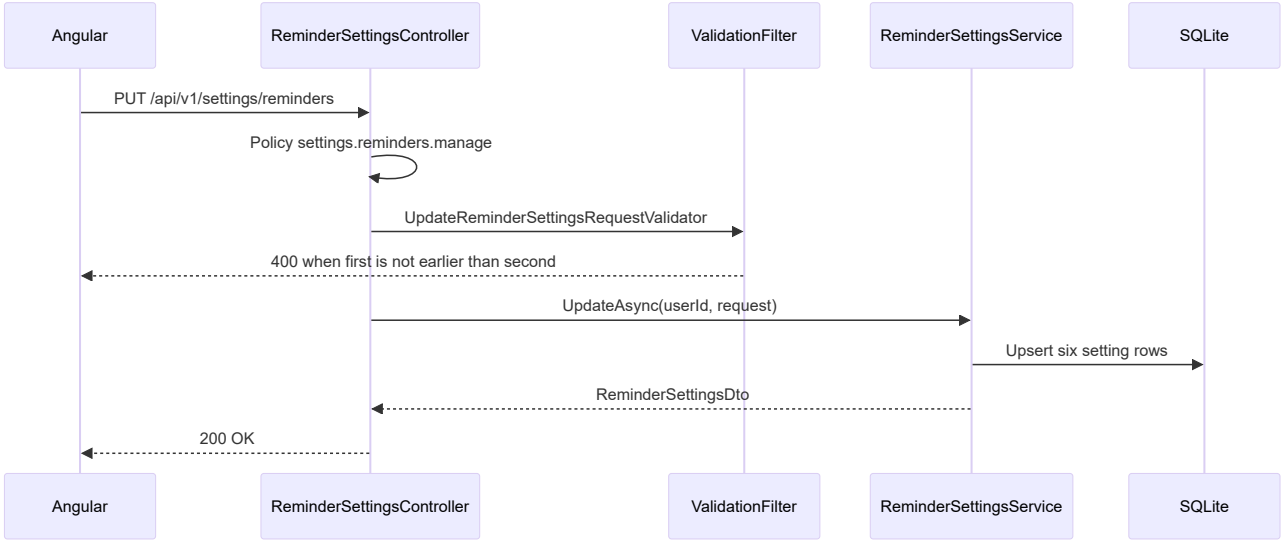
Consumers: `ReminderService` (Communication) skips the whole run when `Enabled` is false. It passes `DeadlineDayEnabled` to `ReminderCalculator.DueReminders` and builds a Mail copy only when `MailEnabled` is true. `TaskService` (Monitoring) reads the defaults for new tasks.

5.4 Repository Layer

Same `IApplicationSettingsRepository` as 08.1 Application Settings:

Method	Description	Query Notes
`GetValuesAsync(keys)`	Values of the given keys	`AsNoTracking`, `ToDictionaryAsync(Key, Value)`
`GetForUpdateAsync(keys)`	Tracked rows for update	Tracked, `ToDictionaryAsync(Key)`
`Add(entity)` (base)	Stage a new row	From `BaseRepository`

5.5 Sequence Diagram



6. Database Design

Table	Column	Type	Constraints	Description
`ApplicationSettings`	`Id`	int	PK, Identity	Row id
`ApplicationSettings`	`Key`	string(128)	Not Null, Unique	This child writes `Reminder.Enabled`, `Reminder.DeadlineDayEnabled`, `Reminder.DefaultFirstDays`, `Reminder.DefaultSecondDays`, `Reminder.DefaultOverdueFrequency`, `Notification.MailEnabled`; reads `Application.Timezone`
`ApplicationSettings`	`Value`	string(1024)	Not Null	Invariant-culture string value
`ApplicationSettings`	`Category`	string(64)	Not Null	`Reminder` or `Notification` for this child
`ApplicationSettings`	`UpdatedByUserId`	int?	– (no FK)	Last editor
`ApplicationSettings`	`UpdatedAt`	DateTime	Not Null	UTC

Relationships: none (key/value table, no FK). The table is shared with 08.1 Application Settings.

Seed data: `SettingsInitializer` inserts missing setting rows at every start and never overwrites existing ones.

> The per-task reminder columns (`FirstReminderDays`, `SecondReminderDays`, `OverdueReminderFrequency` on `Tasks`, migration `20261013104629_AddTaskReminderSettings`) belong to the Monitoring module; this child only supplies their defaults for new tasks.

7. Security & Authorization

- **Authorization:** class-level `[RequirePermission(settings.reminders.manage)]` on `ReminderSettingsController`, so both GET and PUT need it.
- **Data scope:** None. The settings are organization-wide.
- **Frontend:** `permissionGuard` on `settings/reminders` and the menu item in `buildNavigation()`.

Permission Key	Endpoint(s)	UI Element Controlled
`settings.reminders.manage`	`GET`, `PUT /api/v1/settings/reminders`	Menu "Reminder Settings", route `/settings/reminders`

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
Reminder field missing, first not 1–30, second not 0–30	`UpdateReminderSettingsRequestValidator`	400	FluentValidation default
Second reminder not less than first	Validator	400	The first reminder must come earlier than the second reminder.

Shared rules (403, 401, 409 concurrency, 500) are in the parent document. The screen maps the `errors` keys onto the form controls.

9. Integrations / Side Effects

- **Reminders:** `ReminderService` uses `Reminder.Enabled`, `Reminder.DeadlineDayEnabled` and `Notification.MailEnabled` (Mail copy of each reminder). `IAppClock` (08.1) gives the local calendar day.
- **Email / Mail:** No mail is sent by this module. `Notification.MailEnabled` only controls the reminder copies in internal Mail.
- **Tasks:** `TaskService` reads the default first/second days and overdue frequency for new tasks.
- **AI Assistant:** knowledge page `Services/Assistant/Knowledge/reminder-settings.md`.
- **Activity Log:** none (only `UpdatedByUserId` / `UpdatedAt`).

10. Configuration

Key (`appsettings.json`)	Purpose	Default
`Reminders:SchedulerEnabled`, `Reminders:TickSeconds`	Reminder engine that consumes these settings (Communication module)	`true`, `60`
`Application:Timezone`	Fallback for the read-only timezone while the row is missing (owned by 08.1)	`Asia/Kolkata` (code default `UTC`)

Runtime-editable keys in the `ApplicationSettings` table (`SettingKeys`):

- `Reminder.Enabled` (`true`), `Reminder.DeadlineDayEnabled` (`true`), `Reminder.DefaultFirstDays` (`2`), `Reminder.DefaultSecondDays` (`1`), `Reminder.DefaultOverdueFrequency` (`Daily`)
- `Notification.MailEnabled` (`true`)

11. Testing

Test Type	File / Location	Covers
Unit	`tests/TaskFlow.UnitTests/Settings/ReminderSettings/ReminderSettingsServiceTests.cs`	Defaults on a fresh database; missing or unreadable rows fall back to the defaults and the configured timezone; saved values round-trip; save records who and when and leaves the timezone unchanged
Unit	`tests/TaskFlow.UnitTests/Settings/ReminderSettings/ReminderSettingsValidatorTests.cs`	Defaults and a second interval of 0 are valid; a first interval outside 1-30, a second outside 0-30, a second not before the first, an unknown overdue frequency and missing fields or intervals are invalid

Test Type	File / Location	Covers
Integration	`tests/TaskFlow.IntegrationTests/Settings/ReminderSettings/SettingsEndpointsTests.cs`	`GET` works for Manager and Administrator; `PUT` saves and ignores the timezone; invalid values get 400 with field errors and change nothing; missing fields get 400 and do not switch reminders off; Employee gets 403; no token gets 401
Unit (consumer)	`tests/TaskFlow.UnitTests/Communication/Notification/Reminders/ReminderServiceTests.cs`, `ReminderCalculatorTests.cs`	Communication module tests that use these settings: the default 2/1/due-today/daily schedule, weekly overdue, zero or equal intervals, the deadline-day toggle and the application timezone; disabled reminders generate nothing, and mail switched off gives bell notifications only
Frontend unit	`src/app/modules/settings/reminder-settings/pages/reminder-settings.component.spec.ts`	Loads the stored values with a read-only timezone; adds a stored interval that is not a preset option; saves without the timezone and confirms; hides "Saved" after an edit; links field errors to their select; blocks a second interval not shorter than the first; shows server field errors and keeps the form; retryable load error; the Mail delivery toggle
Frontend unit	`src/app/modules/settings/reminder-settings/services/reminder-settings.service.spec.ts`	`GET` and `PUT` on `/api/v1/settings/reminders`; the `PUT` body has no `timezone`

> Shared tests (`SettingsInitializerTests.cs`) are in the parent document.

12. Known Limitations / Gaps found

- The reminder screen offers fixed choices (3/2/1 days and 1/0 days) while the API accepts 1–30 and 0–30.
- **BRD Screen 13 "Email notification setting"**: the code's toggle `Notification.MailEnabled` delivers reminder copies to the internal TaskFlow Mail inbox.
  > ⚠️ To be confirmed: whether any SMTP email channel is tied to this setting (not found in the Settings code).
- **BRD Screen 13 access "Administrator or authorized Manager"** matches the seed defaults (Manager granted `settings.reminders.manage`).
  > ⚠️ To be confirmed: live database grants were not inspected.
- Module-wide limitations (no audit log, no reset to defaults) are in the parent document.

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